

Route G2-X: An Externally Anchored Xenon Feasibility Test

A real response matrix; conditional event yield, not a built source

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Abstract

We conditionally select the uniform X -only Higgs portal and fold the existing driven recoil population through the published XENON1T S2-only nuclear-recoil response. A Higgs-threshold-motivated scale interval is tested against a direct invisible-width constraint and the retained action's vacuum condition. Unknown incident flux remains an explicit axis of feasibility, not an invented experimental capability. A useful negative result follows: throughout the retained recoil-energy window, the scalar scattering kernel factorizes, so this energy-only channel cannot reproduce the old two-line momentum classifier.

1 Physical choices and provenance

This is a conditional phenomenological branch, not a prediction of the radius. Retain $\alpha = .25$, $g = .05$, the selected $m = 0, |j| = 1$ source population, $\sigma_i = .1\Lambda$, and the activated drive $\epsilon = .5$, $\Omega = .2\Lambda$. Choose

$$1 \leq \Lambda = R^{-1} \leq 30 \text{ GeV}, \quad \mu_j = \Lambda \sqrt{25 + j^2}, \quad m_X = \mu_{\pm 1} = \sqrt{26}\Lambda. \quad (1)$$

The range spans the physical Higgs pair threshold $\Lambda_h = m_h/10 = 12.508 \text{ GeV}$ and X_0 masses 5–150 GeV. These boundaries are a coverage choice, not a measured dark mass or a fit to known particles. Use $m_h = 125.08 \text{ GeV}$, $v = 246 \text{ GeV}$, and $\Gamma_h^{\text{SM}} = 4.10 \text{ MeV}$ from the currently served PDG review [1]; Γ_h^{SM} is a theoretical SM prediction. The coupling domain $10^{-5} \leq \lambda_X \leq 1$ is a declared scan range, not a confidence interval or a perturbativity theorem.

The target is natural liquid xenon, using the *actual* official XENON1T 2019 S2-only analysis/2020 response release [2, 3]. We choose the older, directly usable response product instead of calling a newer halo-WIMP exclusion a constraint on this boosted source. Its source commit is 5a364bc8709f2561e5a013ddea6993a5a7c8e313. Both CSV files and their hashes, original documentation and attribution are retained locally. The response includes event-selection losses averaged over the full target. The raw search exposure is

$$\mathcal{E} = 356770 \text{ kg day}. \quad (2)$$

No separate fiducial-efficiency factor is multiplied in. This reuse assumes approximately uniform, thin-target illumination, the source COM frame at rest with respect to the target, and the released energy-only response appropriate to these recoils. It is not a calibration of a new narrow-beam installation.

The released NR matrix extends to hypothetical response below the analysis threshold. We explicitly retain true recoil energy $0.7 \leq E_R \leq 50 \text{ keV}$ and only full S2 bins inside 150–3000 PE. The actual lower retained edge is approximately 150.027 PE; the bin straddling 150 PE is dropped without assuming a sub-bin shape. The response is used as supplied, not replaced by a Gaussian. Energies outside this interval are unmodeled, not asserted to produce zero physical response. No response covariance or new background likelihood is invented.

2 Constraints from the same candidate interaction

At a declared common tree matching prescription, select

$$\Delta\mathcal{L} = -\lambda_X H^\dagger H \sum_j |x_j|^2, \quad \lambda_X \geq 0, \quad \lambda_\Phi = 0. \quad (3)$$

This is a four-dimensional EFT option. The X physical intercept is held fixed by explicit matching,

$$M_{D,0}^2 = 25\Lambda^2 - \lambda_X v^2/2 \geq 0, \quad \lambda_X \leq \lambda_{\text{vac}} = 50\Lambda^2/v^2. \quad (4)$$

With no $|X|^4$ term in the retained action, violating this condition gives a runaway along $H = \Phi = 0$, constant X . Its exclusion is a theory failure, distinct from an experimental exclusion. The analogous claim for a negative Φ bare mass needs its holonomy gradient gap; we do not import the X_0 argument into that sector.

The published ATLAS combination gives $B_{\text{inv}} < .107$ at 95% CL, assuming SM Higgs production [4]. With unchanged visible couplings and all produced dark modes invisible, the permitted new width is

$$\Gamma_{\text{max}} = \frac{.107}{1 - .107} \Gamma_h^{\text{SM}}, \quad (5)$$

$$\Gamma_X = \frac{\lambda_X^2 v^2}{16\pi m_h} B_X(\Lambda), \quad B_X = \sum_{j: 2\mu_j < m_h} \sqrt{1 - 4\mu_j^2/m_h^2}, \quad (6)$$

$$\lambda_X \leq \lambda_h(\Lambda) = \sqrt{\frac{16\pi m_h \Gamma_{\text{max}}}{v^2 B_X(\Lambda)}} \quad (B_X > 0). \quad (7)$$

Every signed label is one complex field with a distinct particle–antiparticle final state. Above Λ_h this particular on-shell decay supplies no bound, not a declaration that all collider constraints disappear. The map labels this a conditional tree recast of a verified 2023 input, not the latest exhaustive global exclusion.

At the matched static tree spectrum, spontaneous contact decay is closed. For $X_j \rightarrow X_l + \Phi_n + \bar{\Phi}_m$ with $j = l + n - m$, the triangle inequality gives

$$\mu_l + m_n + m_m \geq \sqrt{(M_D + 2M_\Phi)^2 + j^2 \Lambda^2} > \mu_j. \quad (8)$$

The particle/antiparticle holonomies cancel. A uniform Higgs cannot change j . Collider escape additionally requires negligible material scattering; it is an explicit recast assumption. The collider is not placed in the hypothetical pump: this constraint uses the drive-off vacuum, while the source forecast retains the drive.

Mixed $g\lambda_X$ loops permit a Φ portal counterterm. Setting it to zero at matching is not an all-orders protection. The cutoff must control the Higgs energy and the modes retained in each application; neither a KK species-count heuristic nor small four-dimensional couplings proves that control. We keep this bounded tree feasibility test conditional, rather than initiate another full matching program.

3 Finite-speed nuclear recoil and the measured response

Use natural xenon number fractions from CIAAW [7], molar mass $\bar{M} = 131.293$ g/mol, and the explicit approximation $m_A = A \times .93149410242$ GeV. Let $m_N = .939$ GeV and $f_N = .308 \pm .018$ [5]; its effective two-nucleon contribution is not added a second time. We approximate the coherent nuclear scalar current by its static normalization and the Helm form factor [6],

$$F_A(q) = \frac{3j_1(qr_A)}{qr_A} e^{-(qs_0)^2/2}, \quad r_A^2 = c_A^2 + 7\pi^2 a^2/3 - 5s_0^2, \quad (9)$$

with $c_A = 1.23A^{1/3} - .60$ fm, $a = .52$ fm and $s_0 = .9$ fm. Arguments use $\hbar c = .1973269804$ GeV fm. This is not a complete finite- q nuclear many-body calculation. The admitted window has $q \lesssim .113$ GeV; no coherent formula is extrapolated to the much larger full recoil endpoints.

For incident momentum p_X in the target rest frame,

$$s = m_X^2 + m_A^2 + 2m_A\sqrt{m_X^2 + p_X^2}, \quad \mathcal{K}(s, m_X^2, m_A^2) = 4m_A^2 p_X^2, \quad (10)$$

$$|\overline{\mathcal{M}}_A|^2 \simeq \frac{4\lambda_X^2 f_N^2 m_N^2 A^2 m_A^2}{(m_h^2 + 2m_A E_R)^2} F_A^2(\sqrt{2m_A E_R}), \quad (11)$$

$$\frac{d\sigma_A}{dE_R} = \frac{\lambda_X^2 f_N^2 m_N^2 A^2 m_A}{8\pi p_X^2 (m_h^2 + 2m_A E_R)^2} F_A^2(\sqrt{2m_A E_R}) \mathbf{1}_{0 < E_R < E_R^{\max}}, \quad (12)$$

$$E_R^{\max} = \frac{2m_A p_X^2}{s}. \quad (13)$$

The incident kinematics are not replaced by a halo-speed distribution. In the NR, zero-form-factor limit, integrating gives $\sigma_A = \lambda_X^2 f_N^2 m_N^2 A^2 \mu_{XA}^2 / (4\pi m_h^4 m_X^2)$, which supplies a normalization check.

Let $r = (s_j, q, p_i)$ label the existing selected source mixture. Here p_i is the dimensionless G2 variable and $p_X = \Lambda \hat{p}_f(p_i)$; the corresponding physical incoming width is $.1\Lambda$. Then

$$dP_r = Z_{\text{pair}}^{-1} w_{l_{s_j}} f_{.1}(p_i) K_{l_{s_j}, 0, q}(p_i) dp_i, \quad l_+ = 0, \quad l_- = -2. \quad (14)$$

These are event-weighted incident *flux fractions* at the sensor only if transmission is common; unknown source luminosity and transport are not thereby computed. With $R_b(E_R)$ the released probability of an S2 bin b , including losses, define

$$\bar{\sigma}_b = \sum_A \xi_A \int dP_r \int_{.7 \text{ keV}}^{50 \text{ keV}} R_b(E_R) \frac{d\sigma_A(p_X(r))}{dE_R} dE_R. \quad (15)$$

Use the supplied energy-bin edges with piecewise-constant response and integrate the smooth cross section within each clipped bin. No independent efficiency correction or repeated velocity factor is used. The unit conversion is $1 \text{ GeV}^{-2} = 3.893793721 \cdot 10^{-28} \text{ cm}^2$.

For stationary, spatially uniform incident selected-particle flux φ_X at the detector in $\text{cm}^{-2}\text{s}^{-1}$,

$$N_b = \varphi_X 86400 \mathcal{E} \frac{1000 N_A}{\bar{M}} \bar{\sigma}_b, \quad \varphi_3 = \frac{3}{86400 \mathcal{E} (1000 N_A / \bar{M}) \sum_{b \in \text{ROI}} \bar{\sigma}_b}. \quad (16)$$

Here N_A is Avogadro's constant, not a nuclear mass number. Three expected signal events is a transparent normalization target, *not* a discovery threshold, a 95% sensitivity, or a fit to observed events. The source flux is missing, so data cannot yet yield a two-parameter xenon exclusion in (Λ, λ_X) .

4 Why this channel loses the old sign information

The exact kinematic lower bound over the entire source support is $p_X \geq .8091444644\Lambda$, attained as $p_i \rightarrow 0$ in the $j = +1, q = -1$ branch. It follows from monotonicity of outgoing two-body momentum with available energy. At fixed dimensionless $\hat{p}, \hat{m}$ and nuclear mass, write $\hat{E} = \sqrt{\hat{m}^2 + \hat{p}^2}$ and $D = m_A^2 + 2m_A \hat{E}\Lambda + \hat{m}^2 \Lambda^2$. Then

$$\frac{dE_R^{\max}}{d\Lambda} = \frac{4m_A \hat{p}^2 \Lambda (m_A^2 + m_A \hat{E}\Lambda)}{D^2} > 0. \quad (17)$$

Checking $\Lambda = 1$ for every isotope therefore bounds the whole continuous interval, not just sampled scales. The minimum endpoint is 9.543 MeV, above 50 keV throughout the range. Consequently

the endpoint indicator is always one in the retained window, and the cross section factorizes:

$$\frac{d\sigma_A}{dE_R} = \frac{\lambda_X^2}{p_X^2} H_A(E_R), \quad \bar{\sigma}_{s_j b} = \lambda_X^2 C_{s_j} D_b, \quad (18)$$

$$C_{s_j} = \int_{s_j} \frac{dP_r}{p_X(r)^2}, \quad D_b = \sum_A \xi_A \int_{.7 \text{ keV}}^{50 \text{ keV}} R_b(E_R) H_A(E_R) dE_R. \quad (19)$$

Any common response $R_b(E_R)$ preserves this factorization. Thus

$$P(b|j = +, \text{accepted}) = P(b|j = -, \text{accepted}), \quad P(j|b, \text{accepted}) = \frac{C_j}{C_+ + C_-}. \quad (20)$$

The optimal energy-only classifier is the majority guess, with error $\min(C_+, C_-)/(C_+ + C_-)$. Better S2 resolution cannot restore the missing momentum dependence within this approximation and window. The original 2.20% error was for a directly measured momentum estimator, not a nuclear recoil deposit. The negative result is about this specific readout, not about the existence of the mode conversion. Response leakage from unmodeled energies, a direction-sensitive response, or additional observations require a different calculation.

5 Numerical feasibility, with the source left unknown

Independent source integration gives, with Λ in GeV,

$$(C_+, C_-) = \frac{(.74075482, .12115857)}{\Lambda^2}, \quad P(j = \pm | \text{accepted}) = (.85943069, .14056931). \quad (21)$$

Hence the best S2-only sign error is 14.0569%, exactly the minority prior within this model; the conditional mutual information is zero. The source quadrature retains its existing $z \leq 10$ tail convention; the kinematic endpoint proof does not rely on that numerical truncation. The accepted-window normalization is

$$\bar{\sigma}_{\text{acc}} = 7.99989 \cdot 10^{-39} \frac{\lambda_X^2}{(\Lambda/\text{GeV})^2} \text{ cm}^2, \quad (22)$$

$$N_{\text{acc}} = .00113108535 \frac{\varphi_X}{1 \text{ cm}^{-2}\text{s}^{-1}} \frac{\lambda_X^2}{(\Lambda/\text{GeV})^2}. \quad (23)$$

The unit-coupling, unit-scale point is a normalization convention, not itself an allowed benchmark. For each scale use $\lambda_{\text{max}} = \min(1, \lambda_{\text{vac}}, \lambda_h)$, omitting λ_h when no Higgs channel is open. Table 1 reports the smallest required flux within that conditional domain.

Λ [GeV]	m_X [GeV]	λ_{max}	φ_3 [$\text{cm}^{-2}\text{s}^{-1}$]	Active condition
1	5.099	.000723	$5.07 \cdot 10^9$	Higgs width
5	25.495	.00176	$2.15 \cdot 10^{10}$	Higgs width
10	50.990	.00382	$1.82 \cdot 10^{10}$	Higgs width
12	61.188	.00857	$5.21 \cdot 10^9$	Higgs width
12.508	63.779	.129	$2.48 \cdot 10^7$	Bare-mass condition
30	152.971	.744	$4.32 \cdot 10^6$	Bare-mass condition

Table 1: Three expected recoils, not discovery sensitivity. The narrow turn near $m_h/10$ is the closing on-shell decay threshold, not a measured preferred radius or an optimized viable point.

The quoted f_N uncertainty alone multiplies yields by .8865–1.1203, or required flux by .8926–1.1280. This is not a total error band: nuclear, response, source and EFT uncertainties remain unquantified. The new verifier passes 573/573 checks; all 3056 preceding Route-G checks also pass unchanged (3629 total). These test implementation and declared mathematical identities, not empirical validation of X . The dense mesh in Fig. 1 only renders these analytic formulas; it is not another source or Gaussian-resolution scan.

6 Decision and remaining physical inputs

The accompanying figure separates the conditional ATLAS exclusion, the retained-action vacuum failure, and the flux needed for three accepted recoils. No white or unhatched region is called experimentally viable: other constraints, production, transport and theory errors are not closed. The response-derived result concerns a calibrated component of an instrument, not the source, full sensor system, or sign classifier.

The energy choice also exposes the drive requirement: $\Omega = .2\text{--}6$ GeV corresponds to ordinary frequency $\nu = \Omega/h \simeq 4.84 \cdot 10^{22}\text{--}1.45 \cdot 10^{24}$ Hz. No physical apparatus that modulates the quartic this way has been specified. It remains a prescribed reservoir, not an ordinary laboratory oscillator; an actual source or a justified quantum energy-exchange process is needed before hardware feasibility can be claimed.

Two productive next choices remain. If only an excess of accepted recoils is required, specify production and at-detector flux, then compare with (16) and an actual background likelihood. If the purpose is to distinguish $j = \pm 1$, change the observable: independent timing/momentum information or a recoil endpoint/directional measurement must be developed. Do not try to recover the missing information by fitting a narrower Gaussian to this S2-only dataset.

References

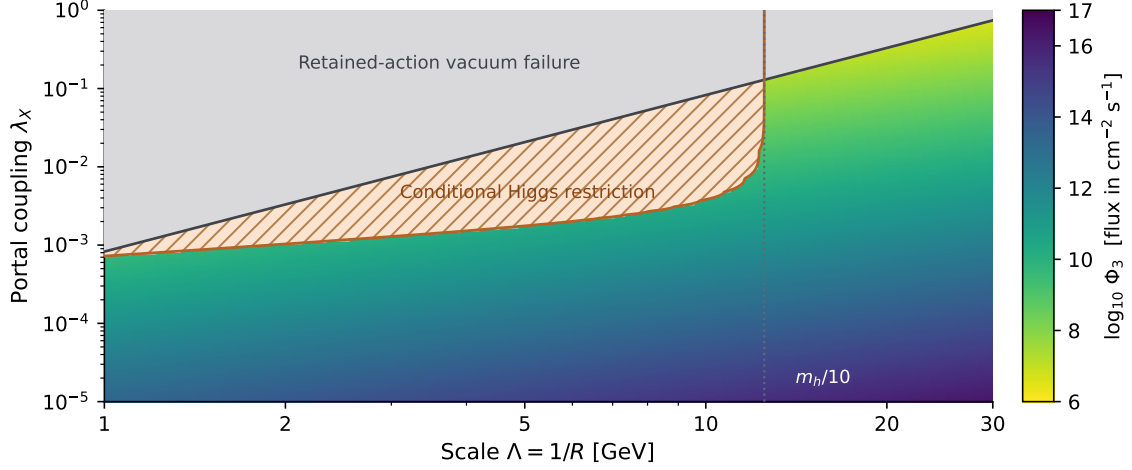
- [1] PDG, *Electroweak Model and Constraints on New Physics*, Sec. 10.4.2, currently served April 2026 revision, review; physical constants.
- [2] XENON Collaboration, *Light Dark Matter Search with Ionization Signals in XENON1T*, Phys. Rev. Lett. 123, 251801 (2019), arXiv:1907.11485.
- [3] XENON Collaboration, *XENON1T S2-only data release* (2020), official repository.
- [4] ATLAS Collaboration, Phys. Lett. B 842, 137963 (2023), arXiv:2301.10731.
- [5] M. Hoferichter et al., Phys. Rev. Lett. 119, 181803 (2017), arXiv:1708.02245.
- [6] J. D. Lewin and P. F. Smith, Astropart. Phys. 6, 87 (1996), Eqs. (4.7), (4.10–11).
- [7] CIAAW, Xenon isotope abundances and standard atomic weight.

Route G2-X | Physical feasibility, not a realized detector

X-only Higgs portal / public XENON1T S2-only NR response

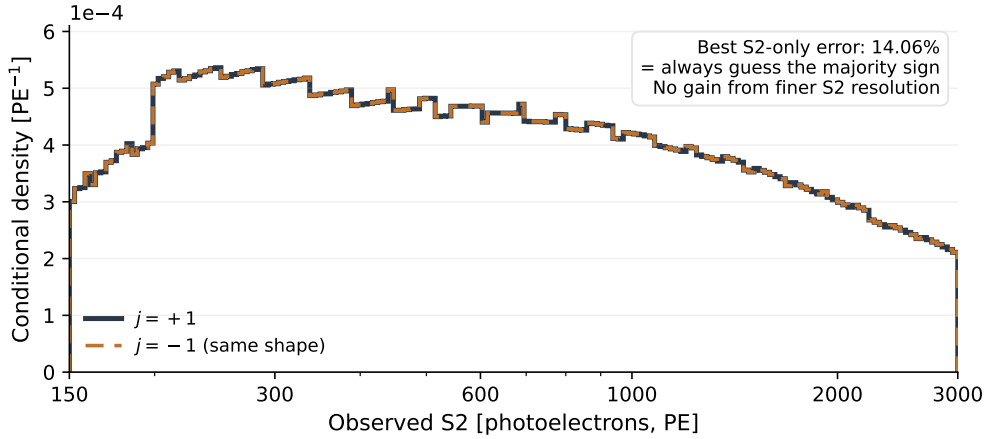
Declared scale: $m_\chi = \sqrt{26} \Lambda$; no measured radius or incident flux

A | Required flux vs conditional restrictions



— Theory: $M_{D,0}^2 < 0$ ATLAS 2023: $B_{\text{inv}} < 0.107$ (95% CL)

B | Real response does not distinguish the two signs



Color = at-target flux for 3 expected accepted events, NOT a discovery threshold.
 No colored region is certified viable: production, transport and backgrounds are missing.
 NR window 0.7–50 keV; full S2 bins 150.027–3000 PE; exposure 356770 kg day.
 Sources: XENON1T 2019 / official 2020 response; ATLAS 2023; PDG; CIAAW.
 Scalar-current/Helm tree model; calibration-informed response, not new measured events.

Figure 1: Conditional feasibility and the sign-readout obstruction. Color encodes a required, not a supplied, flux. Panel B uses the unchanged public response matrix and retains its bin structure.